Simulation and Analysis of Schmitt Trigger

VLSI SYSTEMS DESIGN COURSE PROJECT

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AIM:

To design the transistor-level schematic and create a symbol view for a CMOS Schmitt Trigger in Cadence Virtuoso, which can later be instantiated in higher-level circuits (like digital systems or waveform generators).

APPARATUS/ SOFTWARE REQUIRED:

- Cadence Virtuoso
- Technology Library - gpdk090
- Components - NMOS and PMOS transistors, DC voltage source, Vpulse source,

THEORY:

We use Schmitt Triggers in digital electronics. It is used for rejecting the noise from the input signals by using the hysteresis curve. To provide accurate and more stable results, it uses two threshold voltages i.e., upper threshold voltage (VUT) and lower threshold voltage (VLT). It is mainly used in signal conditioning applications to remove noise from signals which is used in digital circuits, particularly mechanical contact bounce in switches.

It takes an analog signal as an input and converts it into a perfect and stable output signal. It does this by applying positive feedback to the non-inverting input of a comparator. The main purpose of the Schmitt trigger is to eliminate noise or jitter in the input signal and produce a stable output. This is especially true when dealing with signals that switch between two voltage levels.

There are two types of Schmitt Triggers, based on the configuration of the circuit and the polarity of the output relative to the input signal we can classify Schmitt triggers into two types - Non-Inverting and Inverting Schmitt Triggers.

1. Non-Inverting Triggers -

   A non-inverting Schmitt trigger provides a high output when the input is above the upper threshold and a low output when the input is below the lower threshold. This configuration uses a feedback resistor but does not invert the logic levels, maintaining the input signal's polarity.
2. Inverting Triggers -

   An inverting Schmitt trigger provides a high output when the input is below the lower threshold and a low output when the input is above the upper threshold. The feedback resistor inverts the logic levels, creating a hysteresis loop that ensures clean transitions.

CMOS Schmitt triggers are integrated circuits that use complementary metal-oxide-semiconductor technology. They offer low power consumption and high noise immunity, making them suitable for battery-powered and portable applications. CMOS technology ensures that the Schmitt triggers can operate at very low voltages and consume minimal power.

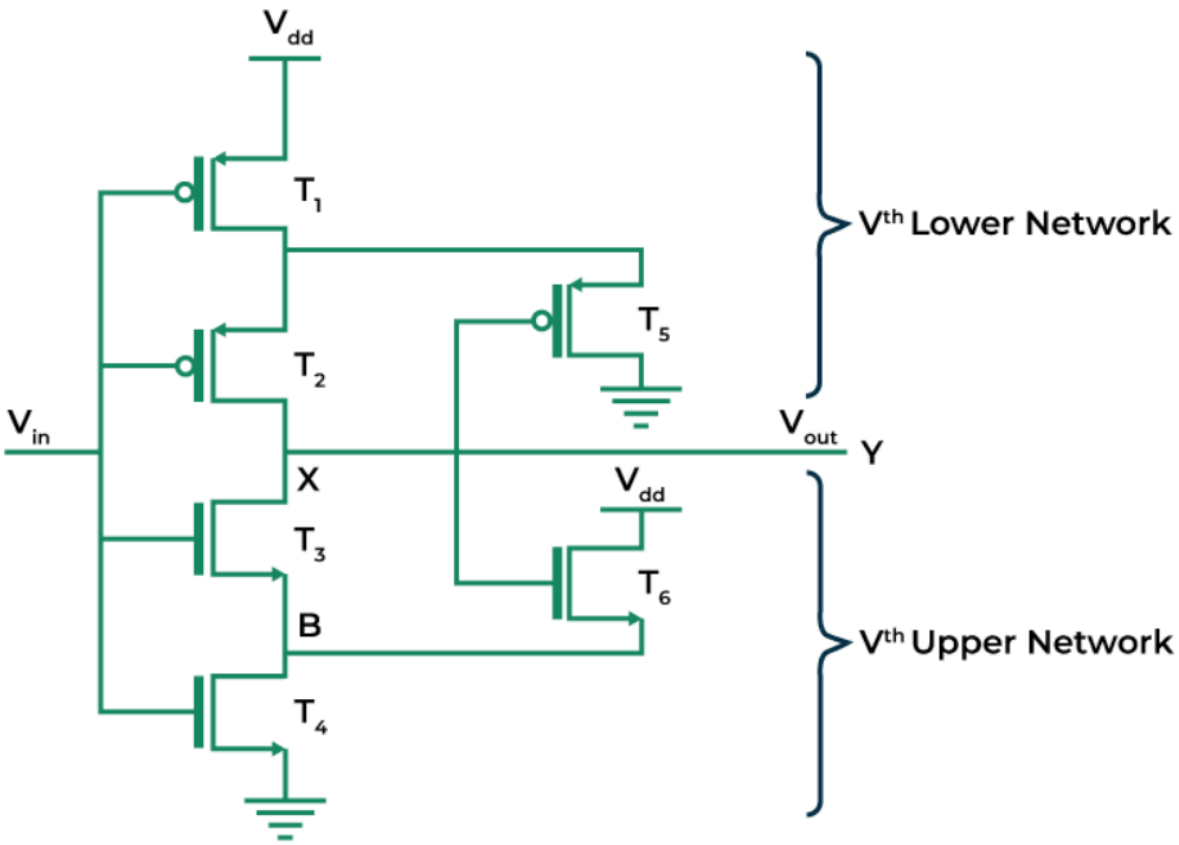
Advantages of CMOS Schmitt Trigger

- Provides noise immunity on slowly varying signals.
- Converts analog signals to digital levels cleanly.
- Prevents multiple unwanted transitions near the threshold.
- Low power consumption (since it’s CMOS).

Applications

- Signal conditioning before digital circuits.
- Waveform shaping and pulse generation.
- Debouncing mechanical switches.
- Oscillators and waveform generators.

WORKING:



A CMOS (Complementary Metal-Oxide-Semiconductor) Schmitt Trigger is an electronic circuit that uses CMOS to make a Schmitt Trigger with exceptionally low power utilization. It utilizes correlative sets of MOSFET semiconductors transistors to accomplish high noise immunity, making it reasonable for battery-worked gadgets and advanced signal handling applications. CMOS Schmitt Triggers have two threshold levels, guaranteeing perfect and stable advanced signal changes while limiting power utilization.

Circuit design: A typical circuit uses two CMOS inverters in series, with a feedback path to create a two-threshold switching behavior.

Input high: When the input voltage (Vin) is high, the output is low. A feedback path ensures that the input must drop significantly to switch the output back to high.

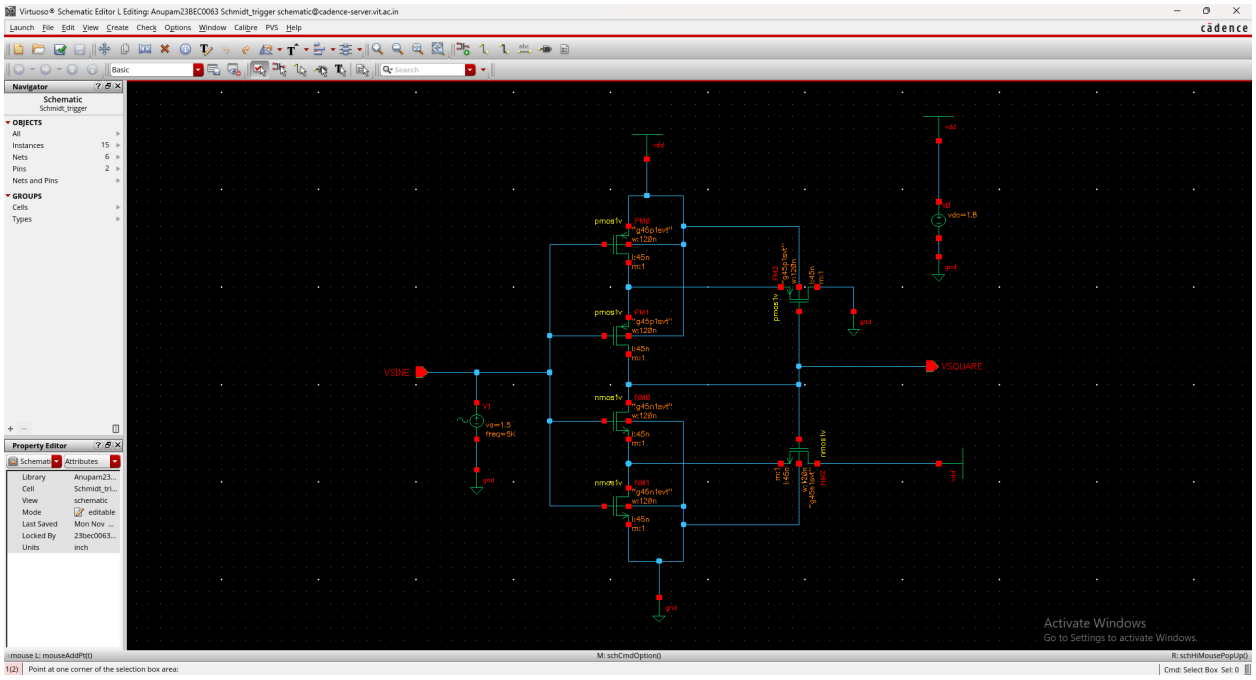
Upper threshold (Vuth): As the input voltage increases, it remains stable until it crosses the upper threshold (Vuth). At this point, a positive feedback mechanism causes the output to switch rapidly from high to low.

Input low: When the input voltage is low, the output is high. The feedback ensures the output stays high even if the input is slightly noisy.

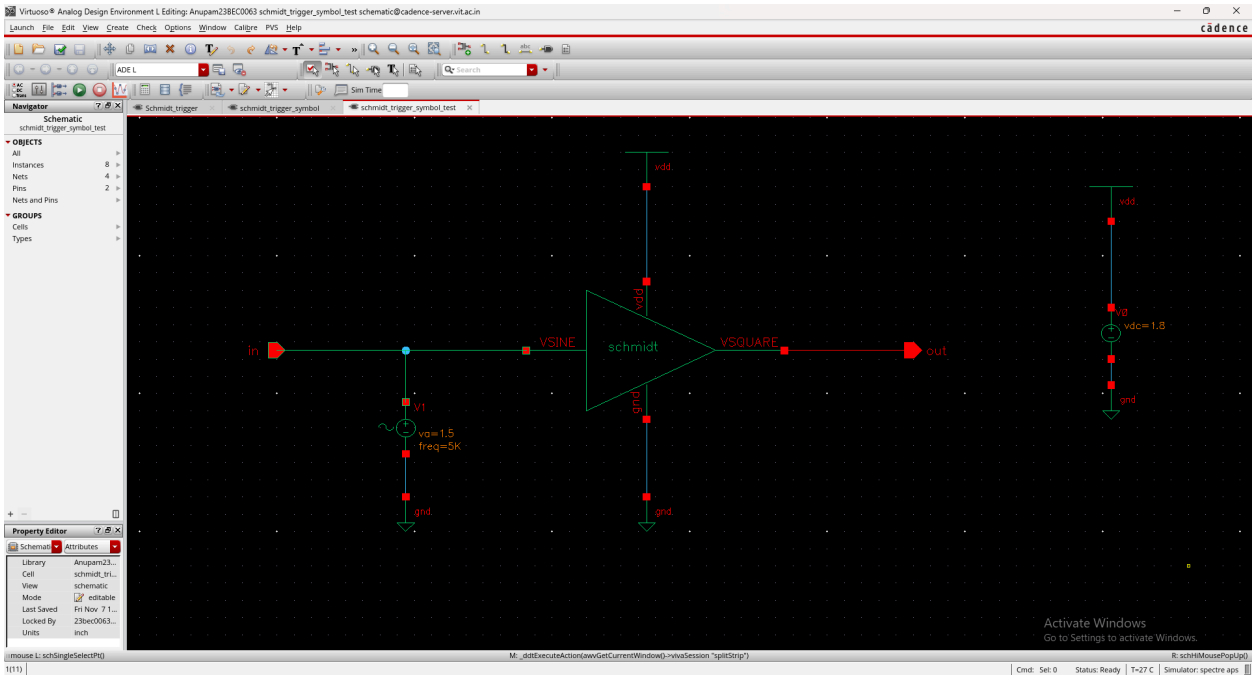
Lower threshold (Vlth): The output stays low until the input voltage drops below the lower threshold (Vlth). At this point, another positive feedback loop switches the output sharply from low to high.

Hysteresis: The difference between (Vuth) and (Vlth) is the hysteresis, which provides noise immunity. This prevents the output from oscillating due to small fluctuations in a noisy input signal.

SCHEMATICS:

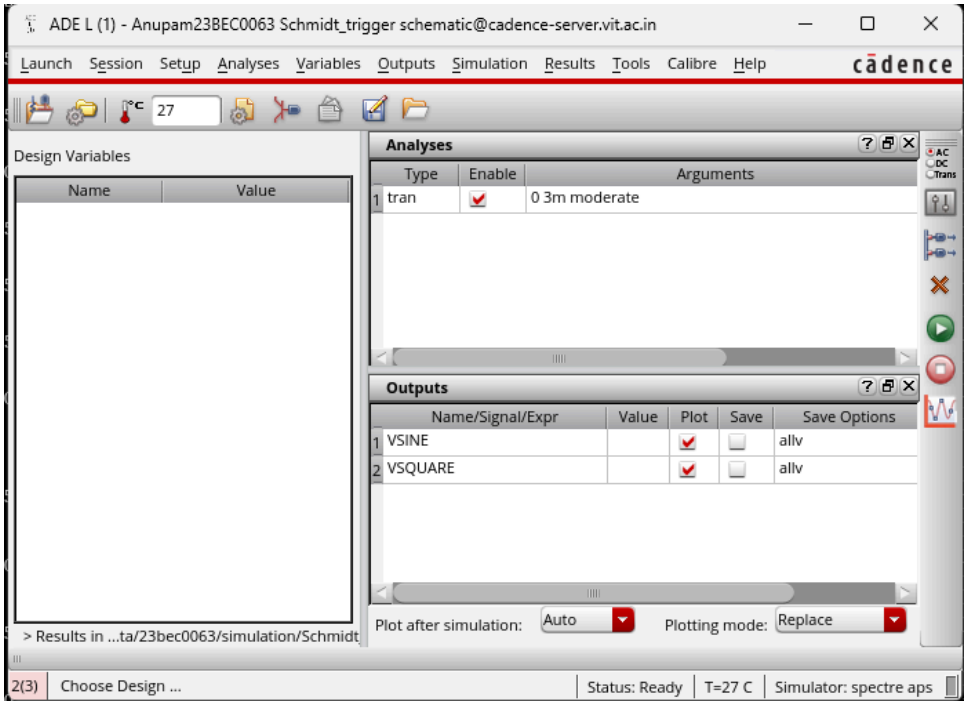


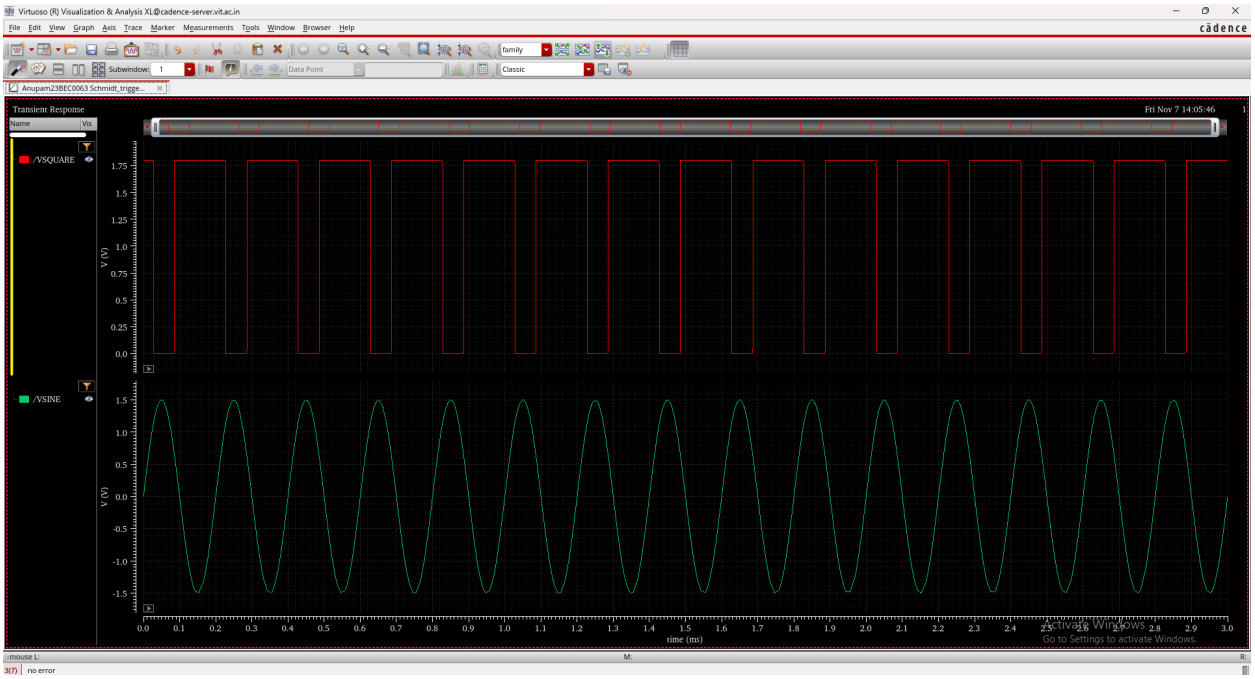
SYMBOL:



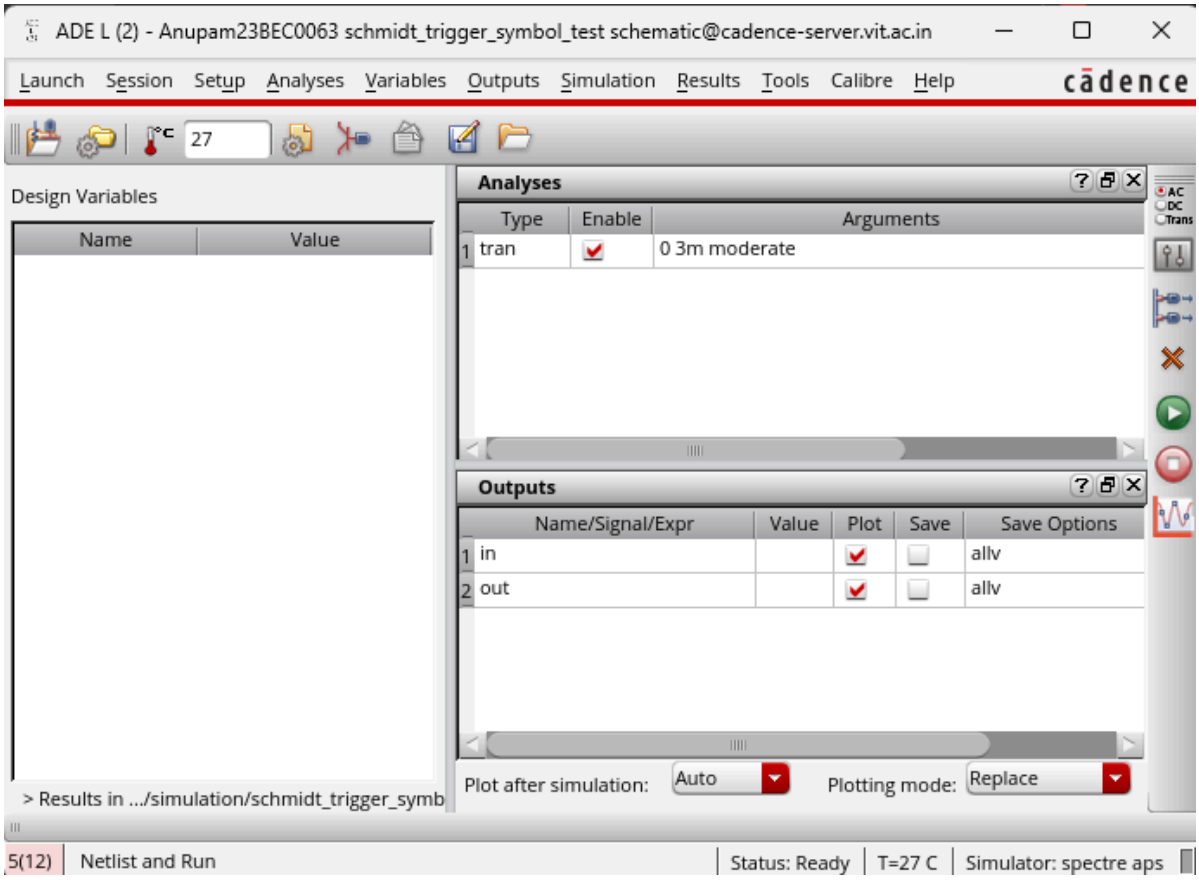
RESULTS AND ANALYTICS:

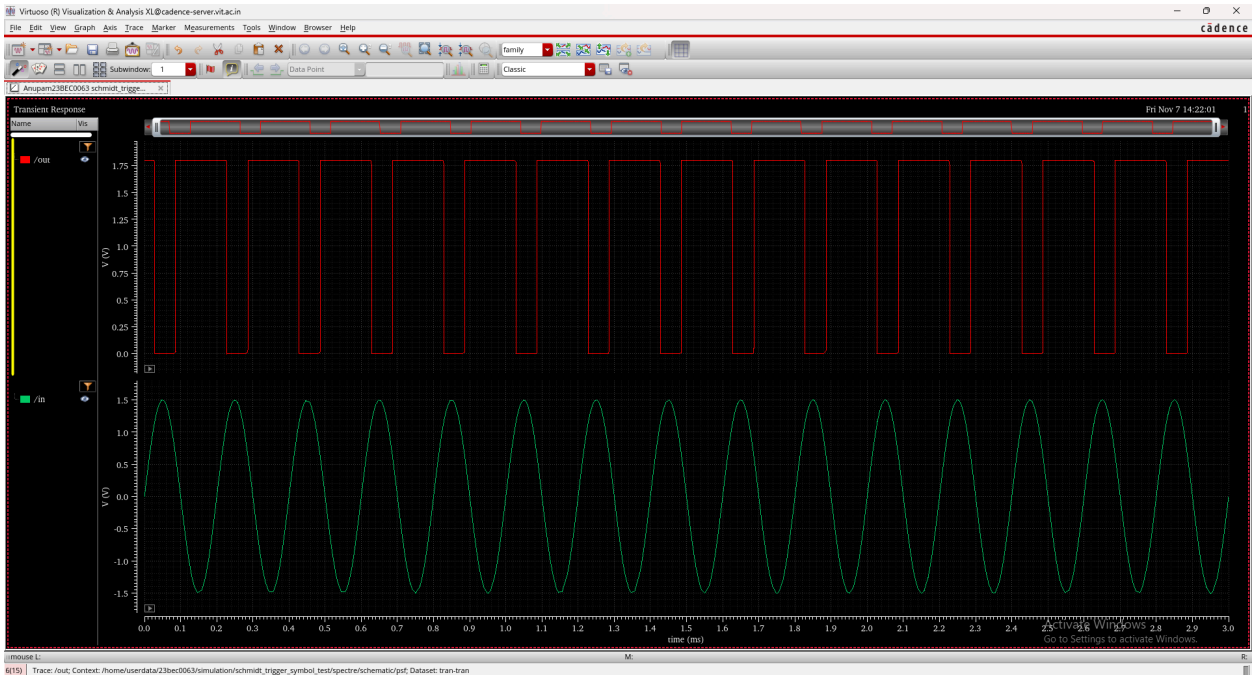
Transient Analysis for Schematic -





Transient Analysis for Symbol -





INFERENCE:

From the simulation of the CMOS Schmitt Trigger in Cadence Virtuoso, it is observed that the circuit successfully exhibits hysteresis behavior, confirming the operation of a noise-immune comparator with two distinct threshold voltages — one for rising input and another for falling input.

The output waveform shows sharp digital transitions even when the input signal varies slowly or contains noise, demonstrating the effectiveness of the positive feedback mechanism in providing stability.

Hence, the CMOS Schmitt Trigger design achieves reliable signal conditioning by suppressing unwanted oscillations and ensuring clean logic-level output transitions.